Introduction to Cloud Reference Architecture

Cloud Architecture:

Cloud architecture is the overall design and structure of a cloud computing system. It shows how different cloud components work together to deliver services over the Internet.

Simple Meaning

Cloud architecture explains:

- What parts make up the cloud
- How they are connected
- How users access cloud services

Main Components of Cloud Architecture:

1. Front End (Client Side)

- **What the user sees and uses**
- **Includes:**
    - Web browser
    - Mobile app
    - Computer or device
- **Example:** Using Google Chrome to access Gmail

2. Back End (Cloud Side)

- **Where all processing happens**
- **Includes:**
    - Servers
    - Storage
    - Databases
    - Virtual machines
    - Cloud software
- Managed by the cloud provider

3. <u>Cloud Services Layer:</u>

- **SaaS** – Software as a Service (Gmail, Zoom)
- **PaaS** – Platform as a Service (Google App Engine)
- **IaaS** – Infrastructure as a Service (AWS EC2)

4. <u>Network / Internet:</u>

- Connects users to cloud services
- Enables data transfer between front end and back end

5. <u>Management & Security</u>:

- **Controls:**
    - Resource allocation
    - Monitoring
    - Backup and recovery
    - Security and access control

<u>Simple Example</u>

When you use Google Drive:

- Your browser = Front End
- Google servers = Back End
- Internet connects them
- Google manages security and storage

Cloud architecture is the blueprint that shows how cloud services are built, connected, and delivered to users over the Internet.

<u>Introduction to reference framework</u>:

When studying **cloud computing**, it is important to use the **correct terminology** to clearly understand how cloud systems are designed and organized. To do this, experts use **reference models, reference architectures, and frameworks**. These act like **guidelines or blueprints** for building cloud systems.

1. <u>Reference Model:</u>

A **reference model** is the **most abstract level**.

- It explains **basic concepts** and how they are related.
- It does **not** give technical details.
- It helps us understand **what components exist** in cloud computing.

Think of it as a **conceptual map**.

2. <u>Reference Architecture:</u>

A **reference architecture** is more **practical**.

- It is built using the ideas from a reference model.
- It shows how **cloud components** can be organized.
- It helps in creating **real cloud systems**.

Think of it as a **blueprint for building** cloud systems.

3. <u>Architecture Framework:</u>

An **architecture framework** provides:

- Rules
- Principles
- Best practices

It explains **how to describe and design architectures** for a specific domain like cloud computing.

- It includes multiple **views or perspectives** (security, users, resources, etc.).

Think of it as a **rulebook** for designing cloud architectures.

4. <u>Generic Reference Framework</u>:

A **generic reference framework**:

- Is defined in a **general way**
- Can be applied to any organization or domain
- Can be customized based on **business needs**

Most **Cloud Computing Reference Architectures (CCRA)** today follow a **common unified model**.

Types of Cloud Computing Reference Architecture Models:

Cloud reference architectures are mainly of **two types**:

1. Role-Based Reference Model

- Focuses on **roles** in cloud computing
- Maps cloud capabilities to roles such as:
  - **Cloud Consumer**
  - **Cloud Provider**
- Helps understand **who does what** in the cloud

Examples:

- **NIST Cloud Computing Reference Architecture (CCRA)**
- **IBM Cloud Computing Reference Architecture (CCRA)**

2. Layer-Based Reference Model

- Organizes cloud services into **layers**
- Each layer has a specific function, such as:
  - Application layer
  - Platform layer
  - Resource layer

Examples:

- Cloud Security Alliance Reference Model
- CISCO Cloud Reference Architecture Framework

In Simple Words

- **Reference Model** → Concepts
- **Reference Architecture** → Structure
- **Framework** → Rules and best practices
- **Role-based** → Who is involved
- **Layer-based** → How services are organized

Actors in Cloud Computing (As Defined by NIST)

Actors in cloud computing are the main parties involved in providing, using, and managing cloud services.

1. Cloud Consumer

Simple Meaning

A **cloud consumer** is a person or organization that uses cloud services.

They are the **main users** of cloud computing.

Explanation:

- Cloud consumers choose cloud services such as **storage, software, or servers**
- They request services from cloud providers
- They pay only for what they use (**Pay As You Go**)
- They sign an **SLA (Service Level Agreement)** with the provider

What SLA Includes

- Response time
- Quality of Service (QoS)
- Performance
- Security
- Availability

Example

- A company using **AWS servers**
- A student using **Google Drive**

A cloud consumer is a person or organization that uses services provided by a cloud provider.

2. Cloud Provider:

Simple Meaning

A **cloud provider** is a company or organization that provides cloud services.

Explanation:

- Owns and manages cloud infrastructure

- Provides services like **IaaS, PaaS, and SaaS**
- Maintains servers, storage, and networks
- Ensures security and service availability

Responsibilities:

- Service deployment
- Resource management
- Cloud service management
- Security and privacy

Examples:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud
- IBM Cloud

A cloud provider is an entity that offers cloud services and manages the required infrastructure.

3. Cloud Auditor

A **cloud auditor** checks whether cloud services are safe, secure, and following standards.

Explanation

- Performs **independent inspection**
- Checks security, privacy, and performance
- Ensures compliance with standards
- Verifies service quality

Example

- A third-party company auditing a cloud provider's security systems

A cloud auditor is an independent party that evaluates cloud services for security, performance, and compliance.

4. Cloud Broker

Simple Meaning

A **cloud broker** acts as a **middleman** between cloud consumers and cloud providers.

Why Cloud Broker is Needed

- Many cloud providers exist
- Choosing and managing services becomes complex
- Broker simplifies service selection and management

Role of Cloud Broker

- Manages cloud services
- Negotiates between consumer and provider
- Adds value to cloud services

Types of Cloud Broker Services

a) Service Intermediation

- Improves cloud services
- Adds features like security, identity management, monitoring

Example:
Providing extra security for cloud access

b) Service Aggregation

- Combines multiple cloud services into one
- Ensures safe data transfer between services

Example:
Using storage from one provider and analytics from another

c) Service Arbitrage

- Selects the best service from multiple providers
- Services are not fixed

Example:
Choosing the cheapest or best-performing cloud service at a given time

A cloud broker manages and negotiates cloud services between cloud consumers and cloud providers.

5. Cloud Carrier

A **cloud carrier** provides the **network connection** between cloud consumers and cloud providers.

Explanation

- Acts as a **transport medium**
- Provides internet or telecom connectivity
- Ensures secure data transmission
- Supports SLA requirements

Examples:

- Internet Service Providers (ISPs)
- Telecom companies

A cloud carrier provides network connectivity and transport of cloud services between cloud consumers and cloud providers.

Quick Summary Table

Actor	Simple Role
Cloud Consumer	Uses cloud services
Cloud Provider	Provides cloud services
Cloud Auditor	Checks security and standards
Cloud Broker	Manages and negotiates services
Cloud Carrier	Provides network connectivity

Architectural Components in NIST Cloud Reference Architecture

NIST defines **five main architectural components** in cloud computing. These components explain **how cloud services are deployed, managed, secured, and delivered**.

1. Service Deployment

Service deployment explains **how cloud services are delivered and accessed** using deployment models and service models.

Cloud Deployment Models (NIST)

Public Cloud

- Services are available to the general public
- Easy access and low cost
- Less secure due to open access

Example: Email services like Gmail

Private Cloud

- Used by a single organization
- High security and control
- Costly to maintain

Example: Bank's internal cloud

Community Cloud

- Shared by organizations with common needs
- Moderate cost and security

Example: Universities sharing research data

Hybrid Cloud

- Combination of public and private cloud
- Critical data in private cloud
- Non-critical work in public cloud

Example: Private cloud for customer data, public cloud for website

Cloud Service Delivery Models

Software as a Service (SaaS)

- Ready-to-use software through the internet

Example: Gmail, Zoom

Platform as a Service (PaaS)

- Platform for developing and deploying applications

Example: Google App Engine, Heroku

Infrastructure as a Service (IaaS)

- Provides servers, storage, and virtual machines

Example: Amazon EC2

2. Service Orchestration

Meaning

Service orchestration is the **coordination and management of cloud resources** to deliver services to users.

Three Layers of Service Orchestration

a) Service Layer

- Allows cloud consumers to access services
- Includes SaaS, PaaS, and IaaS
- Services can work independently or together

"Services can work independently or together"

Services can work independently

Each cloud service model (**SaaS, PaaS, IaaS**) can be used **separately**, without depending on the other services.

Examples:

- **SaaS only** → Gmail, Google Docs
- **PaaS only** → Deploying an app on Heroku
- **IaaS only** → Renting a virtual machine

You can use **any one service alone**.

Services can work together

Different cloud services can also be **combined** to provide a complete solution.

Example 1:

- **IaaS** → Provides virtual machines
- **PaaS** → Runs applications on those machines
- **SaaS** → Delivers the final application to users

Example 2:

- A **SaaS application**
- Uses a **PaaS database**
- Runs on **IaaS infrastructure**

Here, services are **stacked and integrated**.

Services can work independently or together means cloud services like SaaS, PaaS, and IaaS can be used separately or combined to deliver complete cloud solutions.

b) Resource Abstraction and Control Layer

- Converts physical resources into virtual resources
- Uses software like:
    - Hypervisors
    - Virtual machines
    - Virtual storage

1. Virtual Machines (VMs):

Virtual Machines are **software-based computers** that work like real computers.

- They have their own **OS, CPU, memory, and storage**
- Run on a single physical machine
- Multiple VMs can run on one physical server

Example:
Running **Windows and Linux** on the same computer using virtualization.

In simple words: One computer acting like many computers.

2. Hypervisors:

A **Hypervisor** is software that **creates, runs, and manages Virtual Machines**.

- It sits between **hardware and VMs**
- Allocates CPU, memory, and storage to VMs

Types:

- **Type 1 (Bare metal):** Runs directly on hardware
- **Type 2 (Hosted):** Runs on an operating system
  Example: VirtualBox

In simple words: Hypervisor is the **boss** that controls VMs.

3. <u>Virtual Networks and Storage:</u>

<u>Virtual Network:</u>

A **virtual network** is a **software-based network** created from physical network hardware.

- Includes virtual switches, routers, firewalls
- Connects VMs securely

Example:
VMs communicating without using physical cables.

<u>Virtual Storage:</u>

Virtual storage combines physical storage devices into one **logical storage pool**.

- Storage is allocated as needed
- Appears as disks to VMs

Example:
Cloud storage like Google Drive or AWS S3.

One-line:

- **VM:** Software-based computer
- **Hypervisor:** Software that manages VMs
- **Virtual Network & Storage:** Software-based networking and storage

c) <u>Physical Resource Layer:</u>

- Includes actual hardware and facilities
- Servers, storage devices, networks
- Data centres, power, cooling

3. <u>Cloud Service Management:</u>

Cloud service management includes **all activities required to operate and manage cloud services**.

<u>Main Areas of Cloud Service Management</u>

a) <u>Business Support:</u>

- Customer management

- Billing and accounting
- Contract and SLA management
- Reporting and auditing

b) <u>Provisioning and Configuration</u>

- Fast resource allocation
- Resource scaling
- Monitoring and metering usage

c) <u>Portability and Interoperability</u>

- Data transfer between clouds
- Application and VM migration
- Interoperability between cloud services

4. <u>Security</u>

Security ensures **safe and protected use of cloud services**, especially because cloud services are shared and internet-based.

<u>Security Includes</u>

- Authentication and authorization
- Data confidentiality and integrity
- Identity management
- Security monitoring
- Incident response
- Audit and policy management

Security applies to **all layers**, from physical hardware to applications.

5. <u>Privacy</u>

Privacy ensures **protection of users' personal and sensitive information**.

Privacy Responsibilities

- Protect personal and identifiable data
- Secure communication channels

- Ensure compliance with privacy laws
- Cloud auditors check privacy compliance

The NIST cloud reference architecture includes five components: service deployment, service orchestration, cloud service management, security, and privacy. Service deployment defines cloud deployment and service models. Service orchestration manages cloud resources through service, abstraction, and physical layers. Cloud service management handles business support and provisioning. Security ensures safe access to cloud services, while privacy protects users' personal data.

Short Summary

- **Service deployment** → How cloud is delivered
- **Service orchestration** → How resources are coordinated
- **Cloud service management** → Business and operations
- **Security** → Protects cloud systems
- **Privacy** → Protects user data

IBM Cloud Computing Reference Architecture (IBM CCRA)

Cloud Reference Architecture

A **cloud reference architecture** is a **standard blueprint** that explains:

- Who uses the cloud
- Who provides the cloud
- How cloud services are created, managed, and delivered

IBM CCRA explains cloud architecture **from a management point of view**.

Roles in IBM Cloud Reference Architecture

IBM CCRA defines **three main roles**:

1. Cloud Service Consumer (CSC):

A **Cloud Service Consumer** is the **user of cloud services**.

Examples:

- Students using Google Drive
- Companies using cloud-based ERP software

Key points:

- Same as **NIST cloud consumer**
- Accesses cloud services through a **self-service portal**
- Uses services like:
    - SaaS (applications)
    - PaaS (platforms)
    - IaaS (infrastructure)

2. <u>Cloud Service Provider (CSP):</u>

A **Cloud Service Provider** offers cloud services to consumers.

Examples:

- IBM Cloud
- Amazon Web Services (AWS)
- Microsoft Azure

<u>Responsibilities of CSP:</u>

- Build, host, and manage cloud services
- Ensure availability, performance, and security
- Manage billing, monitoring, and service delivery

<u>Admin Portal</u>

- Used by CSP administrators
- Helps in:
    - Publishing services
    - Monitoring performance
    - Managing growth and outages
    - Automating cloud operations
    - Ensuring SLA compliance

Common Cloud Management Platform (CCMP)

CCMP is the **core management system** of IBM CCRA.

Functions of CCMP:

- Delivers cloud services to consumers
- Allows self-service access

Allows self-service access" — what it means

In **CCMP (Common Cloud Management Platform)**, **self-service access** means:

Cloud users can use and manage cloud services by themselves, without asking the provider every time.

In simple words

Users can **select, start, stop, upgrade, or remove cloud services on their own** using an online portal.

No phone calls
No emails
No manual approval for small tasks

Examples

- A company employee creates a **virtual machine** by clicking a button
- A user increases **storage space** instantly
- A business subscribes to a **cloud application** through a web portal
- A user checks **usage and billing details** online

Why self-service access is important

- Saves time
- Faster service delivery
- Reduces workload of cloud provider staff
- Easy and user-friendly

Allows self-service access means cloud consumers can independently request, configure, and manage cloud services through a portal without direct provider intervention.

- Manages services from provider side

CCMP has **two main parts**:

1. **Business Support Services (BSS)**
2. **Operational Support Services (OSS)**

1.Business Support Services (BSS)

BSS handles **business-related activities**.

Main functions:

- Customer management
- Pricing and billing
- Invoicing and accounting
- Order management
- Contract and agreement management
- Service catalogue management

In simple:

BSS manages **money, customers, and business processes**.

2.Operational Support Services (OSS)

OSS handles **technical and operational activities**.

Main functions:

- Service provisioning
- Monitoring and event management
- Incident and problem management
- Virtualization management
- Capacity and performance management
- Change and configuration management

In simple:
OSS manages **technology, infrastructure, and operations**.

3. Cloud Service Creator

A **Cloud Service Creator** develops cloud services.

Responsibilities:

- Designs and develops cloud services
- Uses service creation tools
- Creates service templates
- Integrates BSS and OSS functionalities

Example:

- A developer creating a cloud-based payroll system

Cloud Services in IBM CCRA

IBM CCRA supports:

- **IaaS** – Infrastructure as a Service
- **PaaS** – Platform as a Service
- **SaaS** – Software as a Service
- **BPaaS** – Business Process as a Service

BPaaS (Special feature of IBM CCRA)

- Provides **complete business processes** as services
- Example:
    - Payroll processing
    - Order processing

Payroll processing – Means

Payroll processing means handling employee salary-related work automatically as a cloud service.

It includes:

- Calculating employee salaries
- Deducting taxes, PF, insurance, etc.
- Generating payslips
- Transferring salaries to bank accounts
- Maintaining payroll records

In BPaaS, the company does not do this work itself.

The **cloud provider manages the whole payroll process**.

Simple example

A company uses a cloud service that **calculates salaries and pays employees every month automatically**.

Payroll processing means managing employee salary calculation, deductions, and payments as a cloud-based business service.

Order processing – Means

Order processing means managing customer orders from start to finish as a cloud service.

It includes:

- Receiving customer orders
- Checking product availability
- Generating invoices
- Packing and shipping products
- Updating order status

In BPaaS, this **entire process runs on the cloud**.

Simple example

An online store uses a cloud service that **accepts orders, generates bills, and tracks delivery automatically**.

Order processing means handling customer orders, billing, and delivery through a cloud-based business process service.

Virtualized Infrastructure

Includes:

- Servers
- Storage
- Networks
- Data centres

- Hypervisors (virtualization software)

Note:

- Virtualization management software is part of **OSS**, not infrastructure.

Security, Resiliency, Performance & Consumability:

These are **non-functional requirements** that apply across:

- Cloud services
- Cloud management platform
- Hardware infrastructure

They ensure:

- Data security
- High availability
- Disaster recovery
- Good performance
- Easy usage

Layer-based Cloud Computing Reference Architectures

A **Cloud Reference Architecture (CRA)** is a **layered blueprint** that explains **how cloud systems are designed, managed, and delivered**.
This chapter mainly discusses **two popular layer-based models**:

1. **IETF Cloud Reference Framework (IETF CRF)**
2. **CISCO Cloud Reference Architecture Framework**

1. IETF Cloud Reference Framework (IETF CRF)

The **IETF CRF** divides cloud architecture into **four horizontal layers** and **one vertical layer**.
Each layer has a **specific responsibility** and works together with other layers.

Horizontal Layers in IETF CRF

1. Application / Service Layer (ASL)

This is the **topmost layer**, visible to users.

What it does:

- Provides cloud services to users
- Includes **SaaS, PaaS, and IaaS**
- Users access services through a **cloud portal**

Examples:

- SaaS: Gmail, payroll apps
- PaaS: Development and test environments
- IaaS: Virtual machines, databases, networks

Simple meaning:
This layer delivers **actual cloud services** to consumers.

2. Resource Control Layer (RCL)

This layer **manages virtual resources**.

Main functions:

- Resource scheduling
- Resource availability
- Authentication and authorization
- Inter-cloud resource control

Simple meaning:
This layer **decides who gets which resources, when, and how securely**.

Inter-cloud Resource Control means:

Managing and sharing cloud resources between different cloud providers.

Easy meaning:

It is the ability of **one cloud to communicate, coordinate, and use resources from another cloud** when needed.

In simple words:

If **one cloud does not have enough resources**, it can:

- Request resources from **another cloud**
- Share workloads
- Balance load between clouds
-

3. Resource Abstraction and Virtualization Layer (RAVL)

This layer **creates virtual resources** from physical ones.

Key concepts:

- Virtual Machines (VMs)
- Hypervisors
- Virtual networks and storage

Functions:

- Converts physical resources into virtual resources
- Pools resources and allocates them as per demand

meaning:
This layer **hides physical hardware and provides virtual resources to users**.

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Virtual Machines are **software-based computers** that work like real computers.

- They have their own **OS, CPU, memory, and storage**
- Run on a single physical machine
- Multiple VMs can run on one physical server

Example:
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In simple words: Hypervisor is the **boss** that controls VMs.

3. <u>Virtual Networks and Storage</u>

Virtual Network

A **virtual network** is a **software-based network** created from physical network hardware.

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Example:
VMs communicating without using physical cables.

<u>Virtual Storage</u>

Virtual storage combines physical storage devices into one **logical storage pool**.

- Storage is allocated as needed
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Example:
Cloud storage like Google Drive or AWS S3.

Summary:

- **VM:** Software-based computer
- **Hypervisor:** Software that manages VMs
- **Virtual Network & Storage:** Software-based networking and storage

4. <u>Physical Resource Layer (PRL)</u>

This is the **lowest layer**.

Includes:

- Servers (CPU, memory)
- Storage (hard disks)
- Network devices (routers, switches, firewalls)

Simple meaning:
This layer contains the **actual hardware** in data centres.

Vertical Layer – Utility Management:

This layer **cuts across all horizontal layers**.

Functions include:

- Security management
- SLA management
- Configuration management
- Logging, auditing, and monitoring

Simple meaning:
It ensures the cloud system is **secure, reliable, and properly managed**.

Cloud Management Layer (CML)

The **Cloud Management Layer** monitors and controls the entire cloud.

Key functions:

- Automatic deployment
- Real-time monitoring and alerts
- SLA control
- Security enforcement
- Maintenance and upgrades

Four major components of CML:

1. **Cloud Configuration Management (CCM)**
   Maintains consistent cloud configuration throughout its life cycle.
2. **Cloud Service Registry & Audit Management**
   - Registers cloud services
   - Ensures auditing and compliance
3. **Cloud SLA Management (CSM)**
   - Defines, monitors, and enforces SLAs
   - Applies penalties if SLAs are violated

4. **Cloud Service Security Management (CSS)**
    - Authentication and authorization
    - Encryption and access control

2. CISCO Cloud Reference Architecture Framework

Cisco proposes a **service-oriented cloud architecture** focusing on **security, orchestration, and service delivery**.

Layers in CISCO Cloud Framework

1. Data Centre Technology Architecture Layer

- Includes compute, storage, and network
- Hosts all cloud services

Hardware foundation of the cloud

2. Security Architecture Layer

- Provides end-to-end security
- Covers all layers

Security is treated as a core requirement

3. Service Orchestration Layer

- Acts as the **"glue"**
- Maps resources to services
- Uses configuration repositories

Integrates infrastructure into usable services

4. Service Delivery and Management Layer

- Handles provisioning and monitoring
- Manages infrastructure and services

Ensures smooth service delivery

5. Cloud Services Consumer Layer

- User interaction layer

- Accessed through cloud portals

  Where consumers request and manage services

Simple CISCO Use Case Flow

1. User logs into cloud portal
2. Selects available services
3. Requests provisioning
4. Resources are allocated
5. Services are delivered with SLA and security

Summary:

- **IETF CRF** explains cloud architecture using **layered functional design**
- **CISCO CRA** focuses on **service orchestration and secure service delivery**
- Both frameworks help in **designing, managing, and delivering cloud services efficiently**